

MATHEMATICAL STATISTICS

Statistics:

The word 'statistics' which comes from the Latin word "status" meaning a political state, originally meant information useful to the state, for example, information about the sizes of populations and armed forces. Now the word statistics refers to "numerical facts systematically arranged".

Variable:

A characteristic that varies with an individual or an object, is called a variable.

Continuous Variable:

A variable is called continuous variable if it can take an any value fractional or integer within a given domain.

Discrete Variable:

A discrete variable is one that can take only a discrete set of integers that is the values are taken by jumps or breaks.

Frequency Distribution:

The organization of set of data in a table showing the distribution of the data into classes or groups together with the number of observations in each class or group is called a frequency distribution.

Measure of Central Tendency:

There are three main measures of central tendency: the mode, the median and the mean. Each of these measures describes a different indication of the typical or central value in the distribution. The mode is the most commonly occurring value in a distribution.

Following table shows the information of formulae of measure of central tendency.

Number of observations	n
Observations	$X_i, i = 1, 2, 3, \dots, n$
Frequency	$f_i, i = 1, 2, 3, \dots, n$
Arithmetic Mean	$\bar{X} = \frac{\sum f_i X_i}{\sum f_i}$
Geometric Mean	$G = \text{anti log} \left[\frac{1}{n} \sum f_i \log X_i \right]$
Harmonic Mean	$H = \frac{n}{\sum f_i \left(\frac{1}{X_i} \right)}$
Median	$= l + \frac{h}{f} \left(\frac{n}{2} - c \right)$
Mode	$= l + \frac{f_m - f_1}{(f_m - f_1) + (f_m - f_2)} \times h$
Quartile	$Q_k = l + \frac{h}{f} \left(\frac{kn}{4} - C \right), k = 1, 2, 3, 4$

Decile	$D_k = l + \frac{h}{f} \left(\frac{kn}{10} - C \right), k = 1, 2, 3, \dots, 10$
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The following table shows the information of conditions on distribution about their shapes.

Frequency Distribution	Relation
Absolutely symmetrical	Mean = Median = Mode
Asymmetrical or skewed	Mode = 3Median – 2Mean
Extremely skewed or J-shaped	No Relation

Measure of Dispersion:

In statistics, dispersion (also called variability, scatter, or spread) denotes how stretched or squeezed a distribution (theoretical or that underlying a statistical sample) is. Common examples of measures of statistical dispersion are the variance, standard deviation and interquartile range.

The following table shows the information of formulae of measure of dispersion.

Largest Observation	x_m
Smallest Observation	x_0
Range	$R = x_m - x_0$
Coefficient of Dispersion	$\frac{x_m - x_0}{x_m + x_0}$
Quartile Deviation	$Q. D = \frac{Q_3 - Q_1}{2}$
Coefficient of Q. D	$\frac{Q_3 - Q_1}{Q_3 + Q_1}$
Mean Deviation	$M. D = \frac{\sum f_i X_i - \bar{X} }{n}$
Coefficient of M. D	$\frac{M. D}{\text{Mean}} \text{ or } \frac{M. D}{\text{Median}}$
Variance	$S^2 = \frac{\sum f_i (x - \bar{x})^2}{n}$
Standard Deviation	$S = \sqrt{\frac{\sum f_i (x - \bar{x})^2}{n}}$

Coefficient of variance	$\frac{S}{\text{Mean}} \times 100$
Coefficient of S. D	$\frac{S}{\text{Mean}}$

Properties of Variance:

Following are the some properties of variance.

1. $\text{Var}(a) = 0$ where a is a constant
2. $\text{Var}(X + a) = \text{Var}(x)$
3. $\text{Var}(aX) = a^2 \text{Var}(X)$
4. $\text{Var}(X \pm Y) = \text{Var}(X) + \text{Var}(Y)$

Properties of Standard Deviation:

Following are the some properties of standard deviation.

1. S. D $(a) = 0$ where a is a constant
2. S. D $(X + a) = \text{S. D}(X)$
3. S. D $(aX) = |a| \text{S. D}(X)$
4. $\text{S. D}(X \pm Y) = \sqrt{\text{Var}(X) + \text{Var}(Y)}$

Probability:

The numerical evaluation of chance of occurrence of an event is called probability.

Random Experiment:

An experiment which produces different result even though it is repeated a large number of times under essentially similar conditions, is called random experiment.

Sample Space:

A set consisting of all possible outcomes of a random experiment is called sample space.

Event:

A particular outcome of an experiment is called an event.

Mutually Exclusive Events:

Two events of a single experiment are said to be mutually exclusive or disjoint if and only if they cannot both occur at the same time.

Exhaustive Events:

Events are said to be collectively exhaustive, when the union of mutually exclusive events is the entire sample space.

Equally likely Events:

Two events are said to be equally likely, when one events is as likely to occur as the other.

Independent Events:

Two events are said to be independent if the occurrence of one event does not effect the occurrence of other event. Otherwise events are called dependent events.

Following table shows the some information related to probability.

Event	A, B
Sample place	S
Probability of A	$P(A) = \frac{n(A)}{n(S)}$
Probability of A^c	$P(A^c) = 1 - P(A)$
Probability of ϕ	$P(\phi) = 0$
Probability of sub event	if $A \subseteq B$ then $P(A) \leq P(B)$

Following table shows the some information of algebra of probability .

Addition law	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$ If A and B are mutually exclusive then $P(A \cup B) = P(A) + P(B)$ $P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$
Conditional Probability	$P(A/B) = \frac{n(A \cap B)}{n(B)}$
Multiplication of Probabilities	$P(A \cap B) = P(A) P(B/A)$ $= P(B) P(A/B)$ If A and B are independent then $P(A \cap B) = P(A) P(B)$
Bayes' Theorem	$P(A_i / B) = \frac{P(A_i) P(B/A_i)}{\sum_{i=1}^k P(A_i) P(B/A_i)}$ For $i = 1, 2, \dots, k$ Where A_i are mutually exclusive and their union is S and B is any other event of S such that it can occur if one of the A_i occurs.

Distribution:

There are two types of distributions

1. Univariate Distribution:

These are the distributions in which there is only one variable such as the heights of the students of a class marks obtained by the students of a class in a school.

2. Bivariate Distribution:

Distribution involving two discrete variables is called a Bivariate Distribution. For example, the heights and weights of students of a class in a school.

Correlation:

Correlation is a statistical tool which studies the relationship between two variable and correlation analysis involves various methods and techniques used for studying and measuring the extent of the relationship between two variables.

Convariance:

The convariance between two variables x and y with n pairs of observations $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$ is define as

$$\text{cov}(x, y) = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n} = \left(\frac{\sum x_i y_i}{n} - \bar{x} \bar{y} \right).$$

$$\text{Where } \bar{x} = \frac{\sum x_i}{n} \text{ and } \bar{y} = \frac{\sum y_i}{n}$$

Correlation Coefficient:

The number of showing the degree or extent to which x and y are related to each other, is called the correlation coefficient. It is denoted by $p(x, y)$ or r_{xy} or simply r .

Characteristics of Correlation Coefficient:

Folloing are the some characteristics of correlation coefficients.

- (i) $-1 \leq r \leq 1$
- (ii) If $r = -1$, there is perfect negative correlation between x and y , i.e., corresponding to an increase (or decrease) in one variable, there is a proportional decrease (or increase) in the other variable.
- (iii) If $r = 1$, there is perfect positive correlation between x and y , i. e., corresponding to an increase (or decrease) in one variable, there is proportional increase (or decrease) in the other variable.
- (iv) If $r = 0$, then x and y are not correlated, i. e., the changes in one variable are not followed by changes in the other.
- (v) If $0 < r < 1$, there is a positive correlation between x and y , i.e., an increase (or decrease) in one variable corresponds to an increase (or decrease) in the other.
- (vi) If $-1 < r < 0$, there is negative correlation between x and y , i. e., and increase (or decrease) in one variable corresponds to a decrease (or increase) in the other.

Methods of Calculating Correlation Coefficient:

(a) Karl Pearson's Correlation Coefficient:

This formula is applied to the problems in which data can be measured quantitatively.

$$r = \frac{\text{cov}(x, y)}{\sqrt{\text{var}(x) \cdot \text{var}(y)}} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \cdot \sum (y_i - \bar{y})^2}}$$

$$= \frac{\sum (dx_i \cdot dy_i) - \frac{(\sum dx_i)(\sum dy_i)}{n}}{\left[\sum (dx_i)^2 - \frac{(\sum dx_i)^2}{n} \right] \left[\sum (dy_i)^2 - \frac{(\sum dy_i)^2}{n} \right]}$$

Where $dx_i = x_i - A$ and $dy_i = y_i - B$ and A, B are the assumed means of x -series and y -series respectively.

(b) Coefficient of Rank Correlation:

This formula is applied to the problems in which data cannot be measured quantitatively but qualitative assessment is possible such as beauty, honesty etc, in this case the best individual is given rank number 1, next rank 2 and so on. The coefficient of rank correlation is given by the formula.

$$R = 1 - \frac{6 \sum D_i^2}{n(n^2 - 1)}$$

Where D_i is the difference of corresponding rank and n is the number of pairs of observations.

Regression Analysis:

It is the method used for estimating the unknown values of one variable, corresponding to the known values of another variable.

Line of Regression:

The line of regression of y on x gives the best estimate of the value of y for given value of x and is given by

$$y - \bar{y} = b_{yx} (x - \bar{x}); b_{yx} = r \cdot \frac{\sigma_y}{\sigma_x}$$

The line of regression of x on y gives the best estimate of the value of x for given of y and is given by

$$x - \bar{x} = b_{xy} (y - \bar{y}); b_{xy} = r \cdot \frac{\sigma_x}{\sigma_y}$$

Coefficient of Regression:

The coefficient of regression of y on x denoted by b_{yx} and is given by

$$b_{yx} = r \cdot \frac{\sigma_y}{\sigma_x}$$

$$= \frac{\text{cov}(x, y)}{\sigma_x^2} = \frac{n \sum x_i y_i - \sum x_i \sum y_i}{n \sum x_i^2 - (\sum x_i)^2}$$

The coefficient of regression of x on y denoted by b_{xy} and is given by

$$b_{xy} = \frac{r\sigma_x}{\sigma_y} = \frac{\text{cov}(x, y)}{\sigma_y^2} = \frac{n \sum x_i y_i - \sum x_i \sum y_i}{n \sum x_i^2 - (\sum x_i)^2}$$

Properties of Regression Coefficients:

- (i) Both regression coefficient have the same sign, i. e., either both are positive or both are negative.
- (ii) The sign of correlation coefficient is same as that of regression coefficient, i. e., $r > 0$ if $b_{xy} > 0$ and $b_{yx} > 0$; and $r < 0$ if $b_{xy} < 0$ and $b_{yx} < 0$.
- (iii) The coefficient of correlation is the geometric mean between the two regression, coefficients.

$$r = \pm \sqrt{b_{yx} \times b_{xy}}$$

The sign to be taken outside the square root is that of the regression coefficients.

- (iv) Both the regression coefficients cannot be numerically greater than one.
- (v) Regression coefficients are independent of change of origin but not scale.
- (vi) A. M. of the regression coefficients is greater than the correlation coefficient.

Properties of Lines of Regression:

The two lines of regression have the following properties:

- (i) The two lines of regression pass through the point $(\bar{x}, \bar{y})$
- (ii) Slope of the line of regression of y on x = b_{yx} .
- (iii) Slope of the line of regression of x on y = $\frac{1}{b_{xy}}$
- (iv) The angle θ between two lines of regression is given by $\tan \theta = \pm \left(\frac{1-r^2}{r} \right) \frac{\sigma_x \sigma_y}{\sigma_x^2 + \sigma_y^2}$

If $r = 0$, then $\tan \theta = \infty \Rightarrow \theta = \frac{\pi}{2}$ i. e., if two variables are uncorrected, the lines of regression are perpendicular to each other. If $r = \pm 1$, then $\tan \theta = 0 \Rightarrow \theta = 0$ or π , i. e., in the case of perfect correlation ($r = \pm 1$) the two lines of regression coincide.

Multiple Choice Questions:

1. The organization of a set of data in a table showing the distribution of data into classes together with the number of observations in each class is called:
(a) truth table (b) frequency distribution
(c) polygon (d) data table
2. The precise numbers which separate one class from another in a frequency distribution are called:
(a) class limits (b) class boundaries
(c) class mark (d) class width
3. The difference between the largest and smallest value of the data is called:
(a) length (b) distance
(c) range (d) width
4. Which of the following is a graphical representation of data:
(a) bar charts (b) Histogram
(c) pie-chart (d) all of these
5. In a cumulative frequency polygon if cumulative distribution increases only in jumps and is constant between jumps, then variable is:
(a) continuous (b) discrete
(c) constant (d) smooth
6. The tendency of the observations to cluster in the central part of the data set is called:
(a) mean (b) median
(c) mode (d) central tendency
7. Which of the following is a measure of central tendency.
(a) mean (b) median
(c) mode (d) all of these
8. If the single peaked frequency distribution is absolutely symmetrical, then which of the following is correct.
(a) $\text{Mean} > \text{Median} > \text{Mode}$ (b) $\text{Mean} < \text{Median} < \text{Mode}$
(c) $\text{Mean} = \text{Median} = \text{Mode}$ (d) None
9. If $\text{Mean} \neq \text{Median} \neq \text{Mode}$ then frequency distribution is:
(a) symmetrical (b) skewed (asymmetrical)
(c) curved (d) straight
10. If a frequency distribution is neither skewed nor J-shaped then which of the following is correct.
(a) $\text{Mode} = 3 \text{ Median} - 2 \text{ Mean}$ (b) $\text{Mode} = \text{Median} = \text{Mean}$
(c) $\text{Mode} = \text{Median} + \text{Mean}$ (d) None
11. Which of the following is easy to calculate and simple to comprehend.
(a) Arithmetic mean (b) Geometric mean
(c) Harmonic mean (d) All of these
12. Which of the following is a measure of dispersion.
(a) Percentile (b) Quartile
(c) Standard deviation (d) All of these

13. Which of the following is a formula for quartile deviation.
- (a) $Q_3 - Q_1$ (b) $\frac{Q_3 - Q_1}{Q_3 + Q_1}$
- (c) $\frac{Q_3}{Q_3 + Q_1}$ (d) $\frac{Q_4 - Q_2}{Q_3}$
14. Coefficient of mean deviation is:
- (a) $\frac{\text{M.D.}}{\text{Median}}$ (b) $\frac{\text{M.D.}}{\text{Mode}}$
- (c) $\frac{\text{M.D.}}{\text{Range}}$ (d) none
15. The mean of the squares of deviations of all the observations from their mean is called:
- (a) variance (b) standard deviation
- (c) skewness (d) mean deviation
16. The smaller value of the standard deviation indicates that most of observations in the data set are close to the
- (a) mean (b) median
- (c) mode (d) variance
17. Which of the following is a balancing point of data set.
- (a) arithmetic mean (b) standard deviation
- (c) mode (d) variance
18. Which of the following is independent of origin.
- (a) variance (b) median
- (c) mean (d) mode
19. Which of the following is true for variance.
- (a) $\text{Var}(X + a) = \text{Var}(X)$ (b) $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$
- (c) $\text{Var}(aX) = a^2 \text{Var}(X)$ (d) All of these
20. Which of the following is true for standard deviation.
- (a) $\text{S.D}(a) = 0$ (b) $\text{S.D}(X + a) = \text{S.D}(X)$
- (c) $\text{S.D}(X + Y) = \sqrt{\text{Var}(X) + \text{Var}(Y)}$ (d) All of these
21. Which of the following cannot be negative.
- (a) variance (b) mean
- (c) mode (d) standard deviation
22. The power to which deviations are raised before averaging them is called:
- (a) standard deviation (b) moment
- (c) skewness (d) mode
23. Which of the following is rth moment about mean for population data.
- (a) $\mu_r = \frac{1}{N} (x_i - \mu)^r$ (b) $\mu_r = \frac{1}{N} (x_i + \mu)^r$
- (c) $\mu_r = \frac{1}{N} \sqrt{x_i + \mu}$ (d) $\mu_r = \frac{1}{N} (x_i \mu)^r$
24. Any departure from symmetry of the distribution is called:

- (a) skewness (b) peakness
(c) flatness (d) none
25. If the left tail of the distribution is longer than its right tail, then distribution is:
(a) positively skewed (b) negatively skewed
(c) no skewness (d) centrally skewed
26. If mean > median > mode then distribution is:
(a) positively skewed (b) negatively skewed
(c) centrally skewed (d) none
27. The degree of peakness of a unimodal frequency curve is called:
(a) skewness (b) kurtosis
(c) mesokurtosis (d) flatness
28. Set of all outcomes of an experiment is called:
(a) universal set (b) unique set
(c) sample space (d) underlying set
29. A particular outcome of an experiment is called:
(a) point (b) element
(c) event (d) set
30. The numerical evaluation of a chance of occurrence of an event is called:
(a) probability (b) prediction
(c) analysis (d) average
31. The total number of outcomes of tossing of a three coins together is:
(a) 2 (b) 3
(c) 2^3 (d) 2^4
32. The total number of outcomes in the sample space of rolling of a 4 dice together.
(a) 4 (b) 4^6
(c) 6^4 (d) 24
33. Which of the following is true:
(a) $0 \leq P(E) \leq 1$ (b) $P(E^c) = 1 - P(E)$
(c) probability of certain event is one (d) all of these
34. If two events of an experiment cannot occur together, then these events are called:
(a) mutually exclusive events (b) independent events
(c) exhaustive events (d) equally likely events
35. If the union of two mutually exclusive events is the entire sample space then these events are called:
(a) independent events (b) exhaustive events
(c) equally likely events (d) none
36. If the occurrence of one event does not effect the occurrence of other events, then these events are called:
(a) exhaustive events (b) mutually exclusive events
(c) independent events (d) none
37. The probability of getting at least one head in a toss of three coins is:

- (a) $\frac{3}{8}$ (b) $\frac{5}{8}$
(c) $\frac{7}{8}$ (d) $\frac{4}{7}$
38. The probability of getting a double six in a rolling of two fair dice is:
(a) $\frac{1}{36}$ (b) $\frac{1}{18}$
(c) $\frac{1}{9}$ (d) $\frac{5}{12}$
39. The probability of getting a face card from a pack of 52 playing cards is:
(a) $\frac{1}{52}$ (b) $\frac{3}{52}$
(c) $\frac{12}{52}$ (d) $\frac{1}{3}$
40. If A and B are events then $P(A \cup B) + P(A \cap B) =$
(a) $P(A)$ (b) $P(A) P(B)$
(c) $P(A) + P(B)$ (d) $P(A) - P(B)$
41. If $P(A \cup B) = P(A) + P(B)$ then which of the following is true.
(a) A and B are independent events (b) A is a certain event
(c) $P(A) = 1 + P(B)$ (d) None
42. Which of the following is a formula for conditional probability.
(a) $P(A/B) = \frac{P(A \cup B)}{P(B)}$ (b) $P(A/B) = \frac{P(A \cap B)}{P(A)}$
(c) $P(B/A) = \frac{P(A \cap B)}{P(A)}$ (d) $P(B/A) = \frac{P(A \cup B)}{P(B)}$
43. In a tossing of two dice what is the conditional probability that sum is 7 on two dice given that sum is odd.
(a) $\frac{3}{7}$ (b) $\frac{2}{5}$
(c) $\frac{1}{3}$ (d) $\frac{9}{8}$
44. If $P(A \cap B \cap C) = P(A) P(B) P(C)$ then A, B and C are:
(a) mutually exclusive events (b) exhaustive events
(c) independent events (d) dependent events
45. A random variable is discrete if it can assume values which are:
(a) finite (b) infinite
(c) countably infinite (d) (a) & (b) only
46. If a random variable can assume every possible value in an interval $[a, b]$, then random variable is:
(a) discrete (b) continuous
(c) constant (d) free variable
47. A function $f(x) \geq 0$ will be density function if:

- (a) $\int_0^{\infty} f(x) dx = 1$ (b) $\int_{-\infty}^{\infty} f(x) dx = 1$
- (c) $\int_{-\infty}^0 f(x) dx = 1$ (d) none
48. If the random variable X is continuous with probability density function $f(x)$, then $E(X) =$
- (a) $\sum_{i=1}^{\infty} x_i f(x_i)$ (b) $\int_{-\infty}^{\infty} xf(x) dx$
- (c) $\int_{-\infty}^{\infty} |x| f(x) dx$ (d) none
49. The mathematical expectation of the number of heads while tossing of three coins is:
- (a) $\frac{1}{2}$ (b) 1
- (c) 1.5 (d) $\frac{1}{3}$
50. If X and Y are discrete random variables and a and b are constant, then which of the following is true.
- (a) $E(aX + b) = aE(X) + b$ (b) $E(X + Y) = E(X) + E(Y)$
- (c) $E(XY) = E(X)E(Y)$ (d) all of these
51. Which of the following is a discrete probability distribution.
- (a) Binomial distribution (b) Poisson's distribution
- (c) Hypergeometric distribution (d) all of these
52. Which of the following is a continuous probability distribution.
- (a) exponential distribution (b) normal distribution
- (c) gamma distribution (d) all of these
53. In which distribution mean and variance are μ and σ^2 respectively.
- (a) Poisson's distribution (b) Gamma distribution
- (c) Binomial distribution (d) Normal distribution
54. The mean deviation of a normal distribution is approximately.
- (a) $\frac{1}{2}$ (b) 1
- (c) $\frac{4}{5}$ (d) 0
55. In a normal distribution the normal curve has point of inflection which is equidistant from the
- (a) mean (b) median
- (c) mode (d) variance
56. The quantity which investigates the dependence of one variable on one or more variable is called:
- (a) regression (b) correlation

- (c) covariance (d) skewness
57. The least square regression line passes through.
(a) means of data (b) medians of data
(c) modes of data (d) variance of data
58. The sum of the deviations of the observed values from a least square regression line is equal to:
(a) 0 (b) 1
(c) 0.5 (d) 2
59. The measure of the degree to which any two variables vary together is:
(a) regression (b) correlation
(c) covariance (d) variance
60. If r is a correlation coefficient then:
(a) $-1 \leq r \leq 1$ (b) $0 \leq r \leq 1$
(c) $-1 \leq r \leq 0$ (d) $r \geq 1$
61. If r is a correlation coefficient between X and Y , then which of the following is true.
(a) $r_{XY} = r_{YX}$ (b) $r_{XY} = 1 - r_{YX}$
(c) $r_{XY} \geq 0$ (d) none
62. Which correlation measures the degree of relationship between a variable and its estimate from regression equation.
(a) simple correlation (b) multiple correlation
(c) partial correlation (d) none
63. Which of the following is a suitable curve to fit to data if first differences between successive values are approximately constant.
(a) a straight line (b) a second degree parabola
(c) an exponential curve (d) a logistic curve
64. How many components composed a time series?
(a) 1 (b) 2
(c) 3 (d) 4
65. Which of the following is a component of a time series.
(a) secular trend (b) cyclic fluctuations
(c) random variations (d) all of these
66. To measure a secular trend which of the following method is used.
(a) the method of freehand curve (b) the method of semi-averages
(c) the method of least squares (d) all of these
67. The process of assessing the magnitude of a time series variable which it will assume at some future point of time is called:
(a) probability (b) prediction
(c) forecasting (d) correlation

KEY to MCQ of Mathematical Statistics

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